

# Package ‘GDAinference’

June 21, 2026

**Title** Combinatorial Inference in Geometric Data Analysis

**Version** 0.0.0.9000

**Description** Exact combinatorial (permutation) inference tests for Geometric Data Analysis (GDA). The package implements the multidimensional typicality and homogeneity tests introduced by Le Roux, Bienaise and Durand (2019, ‘Combinatorial Inference in Geometric Data Analysis’, Chapman & Hall/CRC). Tests operate on the principal coordinates produced by common GDA methods and integrate natively with the ‘FactoMineR’, ‘GDAtools’ and ‘ade4’ packages, as well as with plain coordinate matrices. Each test returns a p-value from the combinatorial distribution of a Mahalanobis-type statistic, together with a compatibility region. Unlike the chi-squared approximations sometimes used in practice, the tests are exact (exhaustive enumeration when feasible, Monte Carlo otherwise).

**License** GPL-3

**Encoding** UTF-8

**Roxygen** list(markdown = TRUE)

**Depends** R (>= 4.0.0)

**Imports** utils

**Suggests** FactoMineR,  
GDAtools,  
ade4,  
ggplot2,  
knitr,  
questionr,  
rmarkdown,  
testthat (>= 3.0.0)

**Config/testthat/edition** 3

**VignetteBuilder** knitr

**LazyData** true

**URL** [https://github.com/label42/GDA\\_combinatorial\\_inference](https://github.com/label42/GDA_combinatorial_inference)

**BugReports** [https://github.com/label42/GDA\\_combinatorial\\_inference/issues](https://github.com/label42/GDA_combinatorial_inference/issues)

**Config/roxygen2/version** 8.0.0

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|           |                                                        |
|-----------|--------------------------------------------------------|
| get_coord | <i>Extract principal coordinates from a GDA result</i> |
|-----------|--------------------------------------------------------|

---

## Description

get\_coord() is the compatibility layer of the package: it returns a plain numeric matrix of principal coordinates (one row per point, one column per axis) from the result objects of the most common Geometric Data Analysis packages, so that the inference tests can be run uniformly whatever the upstream GDA tool.

## Usage

```
get_coord(x, axes = NULL, ...)

## Default S3 method:
get_coord(x, axes = NULL, ...)

## S3 method for class 'PCA'
get_coord(x, axes = NULL, ...)

## S3 method for class 'MCA'
get_coord(x, axes = NULL, ...)

## S3 method for class 'FAMD'
get_coord(x, axes = NULL, ...)

## S3 method for class 'CA'
get_coord(x, axes = NULL, ...)

## S3 method for class 'speMCA'
get_coord(x, axes = NULL, ...)

## S3 method for class 'csMCA'
get_coord(x, axes = NULL, ...)

## S3 method for class 'bcMCA'
get_coord(x, axes = NULL, ...)

## S3 method for class 'wcMCA'
get_coord(x, axes = NULL, ...)
```

```
## S3 method for class 'stMCA'
get_coord(x, axes = NULL, ...)

## S3 method for class 'multiMCA'
get_coord(x, axes = NULL, ...)

## S3 method for class 'dudi'
get_coord(x, axes = NULL, ...)
```

### Arguments

|                   |                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------|
| <code>x</code>    | A GDA result object or a numeric matrix/data frame of coordinates.                      |
| <code>axes</code> | Integer vector of axes (columns) to keep, or NULL (default) to keep all available axes. |
| <code>...</code>  | Currently unused; for method extensibility.                                             |

### Details

**Formally tested** object types (see the package's integration tests):

- a numeric matrix or data.frame of coordinates;
- **FactoMineR**: PCA and MCA;
- **GDAtools**: speMCA and csMCA;
- **ade4**: dudi.pca and dudi.acm.

Other object types that share the same structure are **expected to work but are not formally tested**: **FactoMineR** FAMD and CA; **GDAtools** bcMCA, wcMCA, stMCA, multiMCA; and other **ade4** dudi objects (e.g. dudi.coa). For an unsupported object you can always extract the principal coordinates yourself and pass them as a matrix.

### Value

A numeric matrix of principal coordinates with row names preserved when available.

### Examples

```
m <- matrix(rnorm(20), ncol = 2, dimnames = list(NULL, c("Dim1", "Dim2")))
get_coord(m)
get_coord(m, axes = 1)
```

---

plot.typicality\_comb    *Plot a combinatorial typicality test*

---

### Description

Displays, on a plane defined by two axes, the reference cloud with its mean point G, the group mean point C, and the  $(1 - \alpha)$  compatibility region: the principal  $d_\alpha$ -ellipse of the reference cloud, translated so as to be centred on C (Le Roux, Bienaise & Durand 2019, Prop. 3.6). When the analysis is two-dimensional the ellipse is exact; for higher-dimensional analyses it is the ellipse of the displayed plane.

**Usage**

```
## S3 method for class 'typicality_comb'
plot(x, axes = c(1, 2), ...)
```

**Arguments**

|      |                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------|
| x    | A typicality_comb object created with keep_geometry = TRUE (the default of <a href="#">typicality_comb()</a> ). |
| axes | Length-2 integer vector giving the two axes (columns of the analysed coordinates) to display. Default c(1, 2).  |
| ...  | Currently ignored.                                                                                              |

**Details**

Requires the **ggplot2** package.

**Value**

A **ggplot2** object (drawn when printed).

**See Also**

[typicality\\_comb\(\)](#)

**Examples**

```
res <- typicality_comb(Target, Target_group)
plot(res)
```

---

```
print.gdainference_test
```

*Print a combinatorial inference test*

---

**Description**

Print a combinatorial inference test

**Usage**

```
## S3 method for class 'gdainference_test'
print(x, digits = 4, ...)
```

```
## S3 method for class 'gdainference_test'
summary(object, ...)
```

**Arguments**

|        |                                                                            |
|--------|----------------------------------------------------------------------------|
| x      | A gdainference_test object (e.g. from <a href="#">typicality_comb()</a> ). |
| digits | Number of significant digits for the statistics.                           |
| ...    | Ignored.                                                                   |
| object | A gdainference_test object.                                                |

**Value**

x, invisibly.

---

```
print.typicality_byvar
```

*Print a by-category typicality table*

---

**Description**

Print a by-category typicality table

**Usage**

```
## S3 method for class 'typicality_byvar'
print(x, digits = 3, ...)
```

**Arguments**

|        |                                                  |
|--------|--------------------------------------------------|
| x      | A typicality_byvar object.                       |
| digits | Number of decimal places for D and the p-values. |
| ...    | Ignored.                                         |

**Value**

x, invisibly.

---

```
Student
```

*Student example*

---

**Description**

The "Student" didactic data set used for the one-dimensional geometric typicality test in Le Roux, Bienaise and Durand (2019, section 4.4): 10 measurements on a single variable.

**Usage**

```
Student
```

**Format**

A data frame with 10 rows and 1 numeric column (X); row names are the subject identifiers s1...s10.

**Source**

Le Roux, B., Bienaise, S. & Durand, J.-L. (2019). *Combinatorial Inference in Geometric Data Analysis*. Chapman & Hall/CRC.

---

|        |                                        |
|--------|----------------------------------------|
| Target | <i>Target example: reference cloud</i> |
|--------|----------------------------------------|

---

### Description

The "Target" didactic data set used throughout Le Roux, Bienaise and Durand (2019). It is a cloud of 10 points in a two-dimensional geometric space, serving as the reference cloud for the combinatorial typicality test (chapter 3) and as the cloud for the geometric typicality test (chapter 4).

### Usage

Target

### Format

A data frame with 10 rows and 2 numeric columns (X, Y); row names are the point identifiers i1...i10.

### Source

Le Roux, B., Bienaise, S. & Durand, J.-L. (2019). *Combinatorial Inference in Geometric Data Analysis*. Chapman & Hall/CRC.

---

|              |                                    |
|--------------|------------------------------------|
| Target_group | <i>Target example: group cloud</i> |
|--------------|------------------------------------|

---

### Description

The four impact points compared with the [Target](#) reference cloud in the combinatorial typicality test of Le Roux, Bienaise and Durand (2019, chapter 3). The book reports a Mahalanobis distance  $D = 0.964$  and a combinatorial p-value of  $9 / 210 = 0.043$ .

### Usage

Target\_group

### Format

A data frame with 4 rows and 2 numeric columns (X, Y); row names are the point identifiers c1...c4.

### Source

Le Roux, B., Bienaise, S. & Durand, J.-L. (2019). *Combinatorial Inference in Geometric Data Analysis*. Chapman & Hall/CRC.

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|                  |                                                                       |
|------------------|-----------------------------------------------------------------------|
| typicality_byvar | <i>Combinatorial typicality test for every category of a variable</i> |
|------------------|-----------------------------------------------------------------------|

---

## Description

Runs `typicality_comb()` for each category of a supplementary / grouping variable, comparing the subcloud of each category with the whole cloud, and collects the results in a tidy table. This is the multidimensional, *exact* counterpart of the per-axis approximation offered by `GDAtools'` `dimtypicality()`.

## Usage

```
typicality_byvar(
  reference,
  variable,
  axes = NULL,
  notable = 0.4,
  alpha = 0.05,
  max_samples = 1e+05,
  seed = NULL,
  ...
)
```

## Arguments

|                                   |                                                                                                                                                                                                                                               |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reference                         | Reference cloud: a numeric matrix/data frame of principal coordinates, or a GDA result object from <b>FactoMineR</b> , <b>GDAtools</b> or <b>ade4</b> (coordinates are extracted with <code>get_coord()</code> ).                             |
| variable                          | The grouping variable: a factor / character vector of length $n$ aligned to the active individuals, or a length-one character giving the name of a variable stored in reference (e.g. a <code>quali.sup</code> variable of a FactoMineR MCA). |
| axes                              | Integer vector of axes on which to run the tests, or NULL (default) for all available axes.                                                                                                                                                   |
| notable, alpha, max_samples, seed | Passed to <code>typicality_comb()</code> .                                                                                                                                                                                                    |
| ...                               | Passed on to <code>get_coord()</code> .                                                                                                                                                                                                       |

## Value

An object of class "typicality\_byvar": a list with `table` (a data frame with one row per category: category, `n_c`, `D`, `p_value`, `notable`, `method`, `n_perm`) and `results` (the list of the underlying `typicality_comb()` objects, named by category). Print it for a formatted table.

## References

Le Roux, B., Benaïse, S. & Durand, J.-L. (2019). *Combinatorial Inference in Geometric Data Analysis*. Chapman & Hall/CRC.

## See Also

`typicality_comb()`

## Examples

```
# Illustration on the Target cloud, grouped by the sign of the 2nd axis:
g <- factor(ifelse(Target$Y >= 0, "upper", "lower"))
typicality_byvar(Target, g)

# From a GDA result, naming a supplementary variable directly:
# typicality_byvar(my_mca, "degree")
```

---

typicality\_comb

Combinatorial typicality test for a mean point

---

## Description

Performs the exact combinatorial (permutation) typicality test of Le Roux, Bienaise and Durand (2019, chapter 3). It assesses whether the mean point of a *group* cloud deviates from the mean point (centre) of a *reference* cloud, using as test statistic the squared Mahalanobis distance  $D^2$  between the two mean points, taken with respect to the covariance structure of the reference cloud.

## Usage

```
typicality_comb(
  reference,
  group,
  level = NULL,
  axes = NULL,
  notable = 0.4,
  alpha = 0.05,
  max_samples = 1e+05,
  seed = NULL,
  keep_perm = FALSE,
  keep_geometry = TRUE,
  ...
)
```

## Arguments

- |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| reference | Reference cloud: a numeric matrix/data frame of principal coordinates (one row per point, one column per axis), or a GDA result object from <b>FactoMineR</b> , <b>GDAtools</b> or <b>ade4</b> (coordinates are extracted with <code>get_coord()</code> ).                                                                                                                                                                                                                                                                                                                                                                             |
| group     | <p>The group cloud whose mean point is compared with the reference centre. One of:</p> <ul style="list-style-type: none"> <li>• a numeric matrix/data frame of group-point coordinates, expressed in the same coordinate system (same axes) as reference;</li> <li>• a logical vector (length = number of reference points) flagging the individuals that make up the group;</li> <li>• an integer or character vector indexing reference rows;</li> <li>• a grouping variable (a factor / character vector of length <math>n</math>, or the name of a variable stored in reference) when <code>level</code> is also given.</li> </ul> |



|               |                                                                                                                                                                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| level         | Optional category of the grouping variable group that defines the group to test (e.g. level = "Master"). When supplied, group is read as a supplementary/grouping variable rather than a direct subset. To test every category of a variable at once, use <code>typicality_byvar()</code> . |
| axes          | Integer vector of axes (columns of the principal coordinates) on which the test is run, or NULL (default) for all available axes. Note that the test is performed in the subspace spanned by axes; to obtain the genuine full-cloud test, retain <i>all</i> principal axes in your GDA.     |
| notable       | Notable-limit for the descriptive magnitude of the deviation (on the Mahalanobis-distance scale). Default 0.4 (the book's rule of thumb).                                                                                                                                                   |
| alpha         | Level of the compatibility region. Default 0.05.                                                                                                                                                                                                                                            |
| max_samples   | Maximum number of samples. Exhaustive enumeration is used when $\binom{n}{n_c} \leq \text{max\_samples}$ , Monte Carlo otherwise. Default 1e5, as in the reference script of Le Roux et al. (2019).                                                                                         |
| seed          | Optional integer seed, used in the Monte Carlo case for reproducibility.                                                                                                                                                                                                                    |
| keep_perm     | Logical; if TRUE, the full vector of permutation statistics is stored in the result (component perm). Default FALSE.                                                                                                                                                                        |
| keep_geometry | Logical; if TRUE (default), the coordinates and summary geometry needed by <code>plot.typicality_comb()</code> are stored in the result (component geometry). Set to FALSE for a lighter object.                                                                                            |
| ...           | Passed on to <code>get_coord()</code> .                                                                                                                                                                                                                                                     |

### Details

The combinatorial distribution of  $D^2$  is obtained by considering every subset of  $n_c$  points drawn from the  $n$  reference points (where  $n_c$  is the size of the group). When the number of such subsets,  $\binom{n}{n_c}$ , does not exceed max\_samples, the exact exhaustive distribution is computed; otherwise max\_samples subsets are drawn at random (Monte Carlo). The (inclusive) p-value is the proportion of subsets whose statistic is greater than or equal to the observed  $d_{obs}^2$ .

### Value

An object of class "typicality\_comb" (inheriting "gdainference\_test"): a list with components including statistic (Mahalanobis distance  $D$ ), statistic2 ( $d_{obs}^2$ ), p\_value, n\_sup, n\_perm, method, dim (dimensionality  $L$ ), n, n\_c, notable and compatibility. Print it for a formatted summary.

### References

Le Roux, B., Benaïse, S. & Durand, J.-L. (2019). *Combinatorial Inference in Geometric Data Analysis*. Chapman & Hall/CRC.

### See Also

`get_coord()`

### Examples

```
# Target example from Le Roux et al. (2019), chapter 3:
# a group of 4 points compared with a reference cloud of 10 points.
res <- typicality_comb(Target, Target_group)
res
res$p_value # 9 / 210 = 0.0428...
```